as we know it. If humanity is lucky enough to survive the new nuclear age and live for another thirty to forty million years, geologists of the future studying ice cores, corals, and rocks will still be able to pinpoint the time when the Kennedy-Khrushchev treaty was signed.

That is what the research done by earth scientists has suggested. Deposits of carbon-14, plutonium-239, and iodine-129 (their half-life is close to 6,000, 24,000, and 16 million years, respectively) will point in the same direction: a series of nuclear explosions shook the planet in the 1950s and 1960s. Isotope deposits were so high during that period that present-day geologists regard them as a "golden spike"—a geological marker created by an event of global proportions that indicates the start of a new age in the earth's history. The "golden spike" period began in the early 1950s, most notably 1954, the year of the Castle Bravo test of an American hydrogen bomb in the Marshall Islands, which went out of control and yielded a destructive force of 15 megatons of TNT, 2.5 times that predicted by its creators. It was one thousand times more powerful than either the Nagasaki or the Hiroshima bomb. In 1961, the Soviets exploded the 58-megaton Tsar Bomba in the Novaia Zemlia archipelago of the Arctic Ocean. The yield exceeded the projected one by "only" 8 megatons.[2]

The 1963 treaty opened the door to subsequent negotiations to control and then reduce nuclear arsenals. In slightly more than twenty years, between May 1972 and January 1993, the American presidents Richard Nixon, Jimmy Carter, Ronald Reagan, and George H. W. Bush, and their Soviet and Russian counterparts, Leonid Brezhnev, Mikhail Gorbachev, and Boris Yeltsin, signed a number of agreements that not only limited but actually reduced missile capabilities and nuclear arsenals by a whopping 80 percent. Unfortunately, the era of nuclear arms control launched so dramatically by the Cuban missile crisis is coming to an end as we witness the unraveling of the old treaties that kept the world safe. A particularly troubling development is the American and Russian withdrawal in 2019 from the Intermediate-Range Nuclear Forces Treaty, negotiated at the end of the Cold War by Presidents Reagan and Gorbachev back in 1987. Both parties announced their final withdrawal from the treaty on August 2, 2019.[3]